Low-Resource Nepali Sign Language Alphabet Recognition on NSL23: An Exploratory Cross-Signer Study of RGB and Hand-Landmark Models

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Abstract

Public Nepali Sign Language (NSL) video is scarce. Accuracies published on other NSL collections sit on incompatible class lists and splits, so they cannot be ranked. Isolated alphabet recognition is explored here on NSL23 after a file audit that leaves 375 original, individually labelled clips from seven signers. Aggregate takes without timestamps and cropped copies of the same originals are removed. Signers S1 and S2, the only two with a complete alphabet, are held out in turn (277 train / 98 evaluation clips). Five models share a 30-epoch, three-seed schedule: frozen ImageNet MobileNetV2 on the middle frame, a CNN–BiLSTM, a CNN–Transformer, a multilayer perceptron on MediaPipe landmarks, and a bidirectional LSTM on the same landmarks. On 196 pooled cross-signer development clips, RGB accuracy is 5.8%±0.8% to 7.0%±2.1% against 2.0% chance for 49 classes. A landmark BiLSTM on wrist-relative xyz reaches 24.8% ± 4.0%, mostly on vowels (66.0% ± 4.0% versus 10.0%±3.9% on consonants). Concatenating 20 inter-joint angles and training that BiLSTM as a prototypical encoder [2, 13] raises the exploratory cross-signer estimate to 59.4% ± 3.3% accuracy and 57.4% ± 3.6% macro-F1, with consonants at 61.8% ± 5.7%. S1/S2 informed these follow-on representation choices, so 59.4% is not an untouched confirmatory test. When the other lighting of the held-out signer is used as a 49-way 1-shot support set, a cross-entropy BiLSTM reaches 76.2%–78.2%. A CPU webcam loop with the prototypical encoder runs at 33–42 frames per second on an Apple M4. The loop is a practice demo on isolated letters.

Keywords: Nepali Sign Language; NSL23; cross-signer development evaluation; hand landmarks; MediaPipe; prototypical networks; joint angles; webcam; low-resource sign recognition

1 Introduction

If the same person appears in training and test, a classifier can latch onto a shirt or the room behind the signer. Holding the signer out asks whether the system has learned the letter. Isolated-sign benchmarks such as MS-ASL, WLASL, and AUTSL treat that split as part of the task [7, 12, 17]. NSL still has no public video set of that size.

NSL23 is a public alphabet collection released under CC BY 4.0. It contains 36 consonants and 13 vowels [15, 16]. The files are easy to overcount. The dataset paper reports 1,205 gesture instances in 630 videos. Some files are one labelled letter. Others are long aggregate takes. Others are cropped copies of an original. If every file, or every reported instance, is treated as an independent clip, the sample inflates and the same performance can leak across a split.

The original goal was a laptop-camera alphabet recognizer. A CNN–Transformer on letterboxed RGB was the first attempt, because MobileNetV2 is small [10] and Transformers are already common for sign sequences [1, 18]. Under a frozen signer-held-out protocol those RGB models sat a few points above chance, while the temporal heads reached training accuracy 1. Extra attention layers would have fit the six training people more tightly without changing what the encoder sees.

MediaPipe Hands returns 21 three-dimensional keypoints [19]. The vector is small enough to classify on CPU, and it discards most of the clothing and background that a frozen ImageNet encoder still sees. NSL letter work has already used landmarks on other image sets [11]. Those papers did not score these original isolated clips with S1 and S2 held out.

The 375 audited clips and the two held-out folds stay fixed. Frozen RGB heads are compared with landmark classifiers; joint angles, prototypical training, a 1-shot enrollment check, and a timed webcam loop are added on the same files. Only two people supply a complete 49-class cross-signer seat, and those same folds informed the follow-on choices. The percentages therefore describe an exploratory development sample rather than an untouched final test. Heuju and Rijal's word-level Transformer [4] and Shrestha et al.'s still-image letter CNN [11] used other files and splits, and they are left off any NSL23 leaderboard. Chamachot and Lertniphonphan's 20-angle few-shot setup is reused as a method [2]. Their static-alphabet scores are a different exam.

2 Related work

2.1 Sign recognition and signer splits

Deep sign recognition usually stacks a spatial encoder, a temporal aggregator, and whatever split the dataset ships with [9]. Continuous systems such as Sign Language Transformers work on sentence-length video and a translation objective [1]. Isolated benchmarks such as MS-ASL, WLASL, and AUTSL classify one sign and often publish a user-independent partition [7, 12, 17]. Those percentages belong to other languages and vocabularies. The useful borrowing is the habit of holding signers out rather than shuffling clips.

2.2 Nepali Sign Language systems

Most published NSL systems use private collections. Heuju and Rijal trained Transformer and CNN–RNN models on word videos with a 70/30 split and reported 55.9% on 71 words versus 89% on a nine-word subset [4]. Shrestha et al. used MediaPipe landmarks and a CNN on their own letter images [11]. Attention and landmarks have both been tried for NSL. Neither paper scores the original isolated NSL23 clips. Model comparisons in this study stay on the same held-out files.

NSL23 is a dataset paper [16]. The recordings were made at Sikkim Manipal Institute of Technology in Sikkim, India. This paper reuses those public files: isolated letters from a campus collection. Claims about rural Nepal rest on later in-country data.

2.3 Lightweight RGB encoders and pose

MobileNetV2 is a compact ImageNet classifier [10]. Pretrained visual features sometimes transfer; how well they transfer depends on the domain gap [6]. Isolated NSL letters are hand shapes, often nearly static, shot at mixed resolutions. An ImageNet encoder is frozen as a controlled baseline, even though ImageNet features are a weak description of these hands. Pose pipelines replace the image with keypoints. MediaPipe Hands is built for on-device tracking [19]. It misses under occlusion, blur, and poor light. Those misses are left in the features rather than swapping detectors after seeing errors.

2.4 Geometric hand features and prototypical classifiers

Wrist-relative coordinates still rotate with the camera and the forearm. Inter-joint angles from the same 21 points are invariant to rotation, translation, and uniform scale. Chamachot and Lertniphonphan use a 20-angle descriptor with a prototypical network on static fingerspelling images [2]. Prototypical networks classify a query by Euclidean distance to a class mean [13]. Both ideas are applied here to NSL23 video with a BiLSTM over 16 frames. Scores on their static alphabets are omitted.

3 Dataset and evaluation protocol

3.1 Release audit

NSL23 version 2 was downloaded from the official Zenodo record [15]. Byte sizes and MD5 digests of the four archives and the dataset-detail document match the record. A file-level manifest retains archive, path, signer token, filename, canonical class, recording descriptors, and exclusion reason.

The audit counts 630 decodable videos: 607 individually labelled files and 23 aggregate recordings. The distribution also contains 234 cropped or preprocessed files. The primary sample is the complement of both exclusions: 375 original, individually labelled clips from S1, S2, S7, S8, S10, S11, and S12 (Figure 1). All 49 alphabet classes appear in that set.

Aggregate recordings were opened and watched. Filenames were ignored as labels. The release has no per-gesture timestamps. Equal-duration bins in published alphabet order drift; at least one sequence appears incomplete or shifted. An equal-duration cut can therefore assign the wrong class while still showing a plausible hand. Those bins are never used as labels.

Cropped copies are excluded so that a derived rendering of the same take cannot sit in the opposite fold, and so that every prediction points at one original file.

Table 1: Audited NSL23 counts used in this study. "Instances" in the source article are not separately labelled clips.

Quantity	Count
Decodable videos in the release	630
Individually labelled files	607
Aggregate recordings (no timestamps)	23
Cropped or preprocessed files	234
Primary original isolated clips	375
Primary signers	7
Alphabet classes	49
Training clips per fold	277
Test clips per fold (S1 or S2)	98
Seeds × models × folds	$3 \times 5 \times 2 = 30$ runs

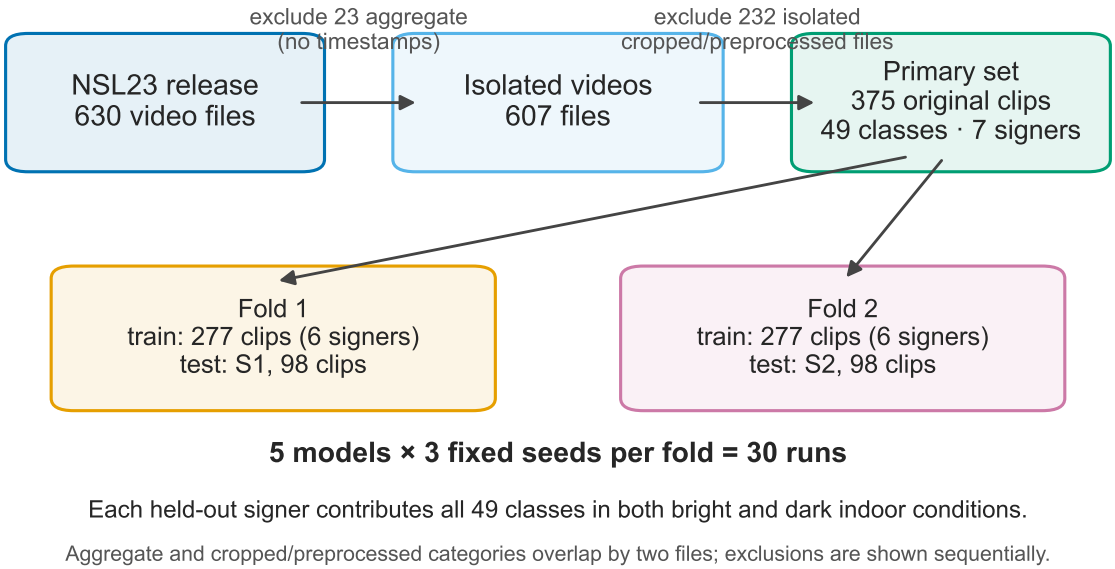


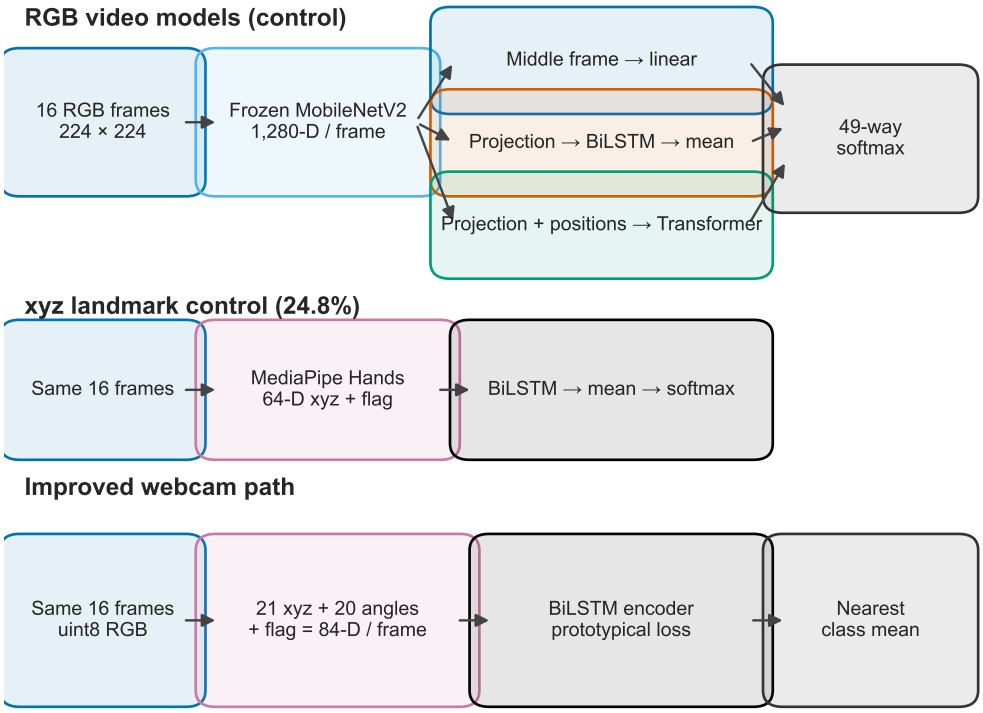
Figure 1: From the NSL23 release to the 30-run matrix. Aggregate files and cropped derivatives are removed before the two signer-held-out folds.

3.2 Class list and folds

The ontology is 36 consonants followed by 13 vowels. Filename aliases are resolved with signer-aware rules because the release uses more than one transliteration. The label set stays at those 49 letters. Every test clip has a class that also appears in training, and each fold contains all 49 labels on both sides of the split.

Only S1 and S2 have original isolated clips for the full alphabet. Fold 1 trains on S2, S7, S8, S10, S11, and S12 and tests on S1. Fold 2 swaps S1 into training and tests on S2. Each test signer contributes one bright and one dark original clip per class. There is no validation set and no early stopping. The epoch-30 checkpoint is evaluated once per run.

The folds are uneven by sign type. S7, S8, S10, S11, and S12 contribute prepared consonants only. Isolated vowels exist only for S1 and S2, so each fold learns every vowel from the



The camera demo uses the 84-D prototypical encoder. Closed 1-shot enrollment replaces train means with the other lighting of the test signer.

Figure 2: RGB controls (top), xyz landmark softmax (middle), and the improved webcam path (bottom): 84-D xyz+angle frames and nearest class prototype.

other complete-alphabet signer. A correct vowel prediction therefore crosses a one-to-one signer boundary. Consonant training has more signers, but the test person and the unprepared indoor setting remain unseen.

3.3 Frame sampling

Sixteen frames are taken at the centres of equal temporal bins of each clip, resized with aspect ratio preserved, and letterboxed to 224×224. ImageNet channel mean and standard deviation are applied at load time. Training uses brightness and contrast perturbations drawn once per clip and applied to all 16 frames. Horizontal flipping is disabled. The single-frame CNN uses the middle element of the same 16-frame sample.

4 Methods

Figure 2 shows the two pipelines. RGB models share a frozen encoder. Landmark models share MediaPipe Hands and a wrist-relative 64-dimensional frame vector.

4.1 Frozen RGB models

Let a clip be $\mathbf{X} \in \mathbb{R}^{T\times 3\times 224\times 224}$ with $T = 16$. A frozen MobileNetV2 encoder f_θ with ImageNet-1K V2 weights maps each frame to $\mathbf{z}_t \in \mathbb{R}^{1280}$ [10]. Encoder weights and batch-normalization running statistics are not updated.

The single-frame CNN classifies the middle frame:

$$\mathbf{h} = f_\theta(\mathbf{X}_{\lfloor T/2 \rfloor}), \qquad \hat{\mathbf{y}} = \mathrm{softmax}(W\,\mathrm{Dropout}(\mathbf{h}) + b). \tag{1}$$

It has 62,769 trainable parameters.

The CNN–BiLSTM projects $\mathbf{z}_t$ to $\mathbf{u}_t \in \mathbb{R}^{256}$, applies a one-layer bidirectional LSTM with 128 units per direction [5], layer-normalizes the outputs, and mean-pools time:

$$\mathbf{u}_t = W_p\mathbf{z}_t + b_p, \qquad [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] = \mathrm{LSTM}(\mathbf{u}_{1:T}), \qquad \bar{\mathbf{h}} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{LN}([\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]). \tag{2}$$

It has 736,305 trainable parameters.

The CNN–Transformer uses the same projection, adds a learned positional table $\mathbf{P} \in \mathbb{R}^{T\times 256}$, and applies two encoder layers with four heads and feed-forward width 512 [18]:

$$\mathbf{H} = \mathrm{Transformer}(\mathbf{U} + \mathbf{P}), \qquad \bar{\mathbf{h}} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{LN}(\mathbf{h}_t). \tag{3}$$

Each head computes $\mathrm{softmax}(QK^\top/\sqrt{d_h})V$ with $d_h = 64$. Encoder dropout is 0 (deterministic scaled-dot-product attention on the training host). Head dropout is 0.2. There is no class token and no causal mask. The model has 1,399,345 trainable parameters.

4.2 Hand-landmark models

MediaPipe Hands [19] is run on the same 16 cached RGB frames (`static_image_mode=True`, complexity 1, at most two hands, detection confidence 0.35). If several hands are returned, the highest-scoring hand is kept.

Let $\mathbf{l}_i \in \mathbb{R}^3$ be the image-normalized coordinates of landmark $i \in \{0, \ldots, 20\}$, with $i = 0$ the wrist and $i = 9$ the middle-finger metacarpophalangeal joint. Wrist-relative, scale-normalized features are

$$\mathbf{p}_i = \mathbf{l}_i - \mathbf{l}_0, \qquad s = \|\mathbf{p}_9^{(x,y)}\|_2, \qquad \tilde{\mathbf{p}}_i = \frac{\mathbf{p}_i}{\max(s, \varepsilon)}, \tag{4}$$

with $\varepsilon = 10^{-6}$. The frame vector concatenates the 63 coordinates and a detection flag $d_t \in \{0, 1\}$:

$$\mathbf{x}_t = \left[\tilde{\mathbf{p}}_0; \ldots; \tilde{\mathbf{p}}_{20}; d_t\right] \in \mathbb{R}^{64}. \tag{5}$$

A miss yields the zero vector, including $d_t = 0$. No class-conditional imputation is used.

On the 375 primary clips (6,000 frames), Hands detected a hand in 5,868 frames (97.8%). No clip had zero detections. Training adds isotropic Gaussian noise with standard deviation 0.02 to the 63 spatial coordinates; the flag is not noised.

The landmark MLP flattens the clip to $\mathbb{R}^{1024}$ and uses hidden widths 256 and 128 with

ReLU and dropout 0.3 (301,617 parameters). The landmark BiLSTM is a one-layer bidirectional LSTM with 64 units per direction, layer normalization, temporal mean pooling, and dropout 0.3 (73,137 parameters):

$$[\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] = \mathrm{LSTM}(\mathbf{x}_{1:T}), \qquad \bar{\mathbf{h}} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{LN}([\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]). \tag{6}$$

4.3 Joint angles and prototypical training

The follow-on matrix keeps the same 16-frame MediaPipe cache. From the wrist-relative points $\tilde{\mathbf{p}}_i$, we compute 15 intra-finger flexion angles (three along each of the five rays from the wrist) and five abduction angles at the wrist, following the 20-angle layout of Chamachot and Lertniphonphan [2]. For unit vectors $\mathbf{u}$ and $\mathbf{v}$ from a pivot joint,

$$\phi(\mathbf{u}, \mathbf{v}) = \arccos\big(\mathrm{clip}(\mathbf{u}^\top \mathbf{v}, -1, 1)\big). \tag{7}$$

A miss ($d_t = 0$) yields a zero angle vector. Concatenating the 63 coordinates, the 20 angles, and the detection flag gives

$$\mathbf{x}_t^{\mathrm{raw+ang}} = \big[\tilde{\mathbf{p}}_0; \ldots; \tilde{\mathbf{p}}_{20}; \boldsymbol{\phi}_t; d_t\big] \in \mathbb{R}^{84}. \tag{8}$$

The BiLSTM input width grows from 64 to 84 (83,377 parameters). Cross-entropy training is otherwise unchanged.

Prototypical training uses the same BiLSTM as an encoder g_ψ that maps a clip to $\bar{\mathbf{h}} \in \mathbb{R}^{128}$. Each epoch runs 40 episodes. An episode draws, for every class, five support clips and two query clips from the training fold, with replacement when a class has fewer than seven training clips. Class prototypes are support means. The loss is softmax over Euclidean distances [13]:

$$\mathbf{c}_c = \frac{1}{|S_c|}\sum_{(\mathbf{X},y)\in S_c} g_\psi(\mathbf{X}), \qquad p(y = c \mid \mathbf{X}) = \frac{\exp(-\|g_\psi(\mathbf{X}) - \mathbf{c}_c\|_2)}{\sum_{c'} \exp(-\|g_\psi(\mathbf{X}) - \mathbf{c}_{c'}\|_2)}. \tag{9}$$

After 40 epochs, test clips are classified by nearest *training*-set prototype. The held-out signer never contributes a prototype in this evaluation. An xyz-only prototypical ablation uses the same encoder and episodes with 64-D frames.

4.4 Cropped-hand CNN and fusion

Full-frame RGB leaves the hand as a small region in a 224×224 letterbox. A follow-on control crops a square around the detected hand from the same 16 cached frames, resizes the crop to 224×224, and trains the frozen MobileNetV2 middle-frame classifier on those crops (62,769 trainable parameters). Fusion averages softmax probabilities from a landmark model and the crop CNN with equal weight, specified before seeing test scores.

4.5 Closed 1-shot enrollment

Each test signer has two original clips per class (bright and dark), which is too few for 5-shot enrollment of that person. The 1-shot protocol takes the other lighting of all 49 classes as support and the remaining 98 clips as queries: bright queries use the 49 dark clips, and the reverse. S1 and S2 have every class, so every prototype comes from that signer. An easier variant replaces only the true class prototype and keeps the other 48 classes as other people's means; that number is omitted because it inflates 1-shot accuracy.

4.6 Loss and optimization

The five models in the primary matrix use inverse-frequency class-weighted cross entropy without label smoothing. For class counts n_c in a training fold,

$$w_c \propto \frac{1}{n_c}, \qquad \frac{1}{C}\sum_{c=1}^{C} w_c = 1, \qquad \mathcal{L} = -\sum_{c=1}^{C} w_c \, y_c \log \hat{y}_c, \tag{10}$$

with $C = 49$. The weights offset the release's two isolated clips per vowel versus up to seven per consonant.

AdamW uses learning rate 3×10^{-4} and weight decay 10^{-4} [8], with cosine annealing over 30 epochs, batch size 8, and seeds 17, 42, and 73. Deterministic algorithms are enabled. RGB models train on Apple M4 MPS; landmark models train on CPU. This controls supported nondeterministic operations on the recorded host, but it does not promise bitwise identity across PyTorch releases, hardware, or devices. The schedule, including the absence of a validation split, was fixed before the RGB matrix and reused for landmarks. Prototypical runs use the same optimizer and seeds for 40 epochs, without class-weighted cross entropy, because the episode already visits every class.

4.7 Webcam loop

The demo classifies a rolling window of live or replayed frames. It keeps the latest $T = 16$ landmark vectors from a camera or video stream, concatenates the 20 joint angles, runs the prototypical BiLSTM, and assigns the nearest stored training-class mean. Class probabilities are a softmax over negated distances, optionally smoothed with an exponential moving average

$$\mathbf{s}_t = \alpha \mathbf{s}_{t-1} + (1 - \alpha)\hat{\mathbf{y}}_t, \qquad \alpha = 0.4. \tag{11}$$

Streaming speed is wall-clock frames processed per second, MediaPipe included. RGB clip throughput after offline decoding is reported separately.

5 Experiments

The primary matrix is five models, two folds, and three seeds (30 runs). On the same folds and seeds the follow-on matrix trains xyz+angle cross-entropy models, prototypical BiLSTMs (xyz and xyz+angle), and the cropped-hand CNN, then scores closed 1-shot enrollment. Reusing

S1/S2 to motivate and compare these follow-on representations makes the complete matrix exploratory; "held out" means absent from a particular training run, not untouched by study-level model development. For the prototypical encoder we also record top-3, top-5, a 36-consonant restriction, and 5-way 1-shot episodes. Checkpoints are never selected on those extra exams. Each evaluation fold has 98 clips (49 classes $\times$ {bright, dark}). Seed-level scores pool the two held-out signers (196 unique clips). Architectures are summarized by the mean and sample standard deviation of the three seed scores. Top-1 accuracy and macro-F1 are the main descriptors [14]. Equal class support makes pooled accuracy the same number as macro-recall.

Each run stores class-stratified clip bootstrap intervals (2,000 replicates). They describe prediction variability on these clips [3]. Two held-out signers are too few for those intervals to stand in for a population of signers.

Bright/dark and consonant/vowel splits were locked before the RGB matrix. Illumination was never randomized as an experimental factor. Architecture differences are seed-matched percentage points. Three seeds are too few for a hypothesis test.

RGB batch-one latency used Apple M4 MPS with device synchronization: model-only forward pass versus original-video decode, preprocess, and inference. Landmark clip latency used CPU: model-only versus MediaPipe on 16 cached frames plus the classifier. The webcam loop was timed on original NSL23 `.MOV` files with the display off.

5.1 Reproducibility boundary

Before the first full primary run, 15 protocol, configuration, manifest, loader, model, training, and launcher files were recorded in a dated SHA-256 snapshot. The original snapshot did not capture its Git revision; a later audit found that every recorded byte and size matches Git commit `0eb1ff2` exactly, now named by the convenience tag `v1-protocol-freeze`. The dated hash snapshot remains the authoritative chronology record. A standalone verifier checks that historical Git tree, while a separate worktree mode intentionally reports later Version 2 edits as drift. Active manifests use repository-relative paths; historical run receipts retain the machine paths recorded when those runs occurred.

6 Results

6.1 RGB controls

Table 2 and Figure 3 summarize pooled scores. The single-frame CNN reaches $5.8\% \pm 0.8\%$ accuracy and $3.5\% \pm 0.2\%$ macro-F1. CNN–BiLSTM is $7.0\% \pm 2.1\%$ / $4.2\% \pm 2.1\%$. CNN–Transformer is $6.3\% \pm 0.8\%$ / $4.4\% \pm 1.3\%$. All three beat 2% chance. A usable 49-way tutor would need to sit in a different accuracy band.

Seed-matched gaps among the RGB models are about one to two points (Transformer minus single-frame: $+0.5 \pm 1.0$ accuracy points). Figure 4 shows why a larger gap was easy to hope for. Both temporal RGB heads reach mean training accuracy 1.00 by the last epoch, while test accuracy stays near 6%. The extra capacity memorizes the six training signers.

Table 2: Pooled unseen-signer performance (196 clips/seed; mean ± SD over three seeds). Chance accuracy for 49 classes is 2.0%.

Model	Acc. (%)	Macro-P (%)	Macro-R (%)	Macro-F1 (%)
Single-frame CNN	5.8 ± 0.8	3.6 ± 0.6	5.8 ± 0.8	3.5 ± 0.2
CNN–BiLSTM	7.0 ± 2.1	5.2 ± 5.0	7.0 ± 2.1	4.2 ± 2.1
CNN–Transformer	6.3 ± 0.8	5.7 ± 1.6	6.3 ± 0.8	4.4 ± 1.3
Landmark MLP	18.0 ± 2.1	10.3 ± 1.5	18.0 ± 2.1	11.1 ± 1.8
Landmark BiLSTM	24.8 ± 4.0	17.1 ± 5.6	24.8 ± 4.0	16.8 ± 3.8

Table 3: Held-out signer accuracy and macro-F1 (98 clips/fold; mean ± SD over three seeds).

Model	S1 acc. (%)	S1 F1 (%)	S2 acc. (%)	S2 F1 (%)
Single-frame CNN	6.5 ± 0.6	2.8 ± 0.9	5.1 ± 1.0	2.9 ± 1.3
CNN–BiLSTM	8.2 ± 2.0	5.0 ± 3.2	5.8 ± 2.4	2.0 ± 1.1
CNN–Transformer	5.1 ± 1.0	3.7 ± 1.3	7.5 ± 1.2	3.0 ± 1.4
Landmark MLP	18.7 ± 3.1	11.6 ± 2.5	17.3 ± 1.0	10.1 ± 0.7
Landmark BiLSTM	25.9 ± 2.6	17.4 ± 2.0	23.8 ± 5.6	15.0 ± 4.8

6.2 Landmark models

The landmark MLP reaches 18.0% ± 2.1% accuracy and 11.1% ± 1.8% macro-F1. The landmark BiLSTM reaches 24.8% ± 4.0% and 16.8% ± 3.8%. Relative to the single-frame CNN, the BiLSTM difference is +19.0 ± 3.7 accuracy points and +13.2 ± 3.8 macro-F1 points on the same 196 clips. Relative to the CNN–Transformer it is +18.5 ± 4.6 and +12.4 ± 5.1 points. The BiLSTM also leads the landmark MLP by +6.8 ± 1.9 accuracy points.

Final-epoch training accuracy is 0.37 ± 0.03 for the MLP and 0.58 ± 0.02 for the landmark BiLSTM. Neither saturates the way the RGB temporal heads do.

Table 3 splits the two test people. Landmark BiLSTM is 25.9% ± 2.6% on S1 and 23.8% ± 5.6% on S2. The ordering versus RGB is the same in both folds.

6.3 Lighting and sign type

Table 4 and Figure 5 split the pre-specified bright and dark clips. Landmark BiLSTM is 27.6% ± 3.5% on bright clips and 22.1% ± 4.6% on dark clips. RGB models stay near 5–7% in both conditions. The dark drop is a description of these files. We did not control illumination.

Table 5 is the split that actually explains the landmark gain. RGB vowel accuracy is about 10% for all three image models, and the single-frame CNN already sits in that band, so the temporal heads add little on vowels. Landmark BiLSTM vowel accuracy is 66.0% ± 4.0%; consonant accuracy is 10.0% ± 3.9%. The MLP shows the same pattern at a lower level (48.1% vowels, 7.2% consonants). Most of the landmark gain is on the 13 vowels. The 36 consonants stay hard after pose normalization.

Figure 6 and Figure 7 show where consonants go. Across the 12 repeated evaluations of each class (two clips × two folds × three seeds), BHA, RA, and TA are never recognized. BHA and RA always map to AH; TA always maps to RI; NA always maps to YAN. Those confusions are stable on this protocol.

Table 4: Descriptive bright versus dark accuracy (98 pooled clips/condition/seed).

Model	Bright acc. (%)	Dark acc. (%)	Bright F1 (%)	Dark F1 (%)
Single-frame CNN	5.8 ± 1.6	5.8 ± 0.6	3.2 ± 1.1	3.0 ± 0.7
CNN–BiLSTM	7.1 ± 1.0	6.8 ± 3.3	3.6 ± 1.6	4.0 ± 3.2
CNN–Transformer	7.1 ± 0.0	5.4 ± 1.6	4.5 ± 0.3	3.0 ± 1.5
Landmark MLP	21.4 ± 2.0	14.6 ± 2.1	12.7 ± 2.5	8.8 ± 1.2
Landmark BiLSTM	27.6 ± 3.5	22.1 ± 4.6	18.4 ± 3.1	14.3 ± 4.1

Table 5: Consonant versus vowel performance on the pooled out-of-fold clips. Accuracy uses all clips in the named sign type; F1 is macro-averaged only over classes supported within that sign-type subset (36 consonants or 13 vowels), not over all 49 classes.

Model	Cons. acc. (%)	Vowel acc. (%)	Cons. F1 (%)	Vowel F1 (%)
Single-frame CNN	4.2 ± 0.7	10.3 ± 2.2	2.8 ± 0.5	10.9 ± 2.6
CNN–BiLSTM	5.6 ± 1.8	10.9 ± 2.9	4.2 ± 1.8	8.8 ± 3.5
CNN–Transformer	4.9 ± 0.7	10.3 ± 2.9	4.0 ± 0.8	9.5 ± 2.6
Landmark MLP	7.2 ± 2.4	48.1 ± 1.9	5.6 ± 2.5	45.0 ± 2.8
Landmark BiLSTM	10.0 ± 3.9	66.0 ± 4.0	8.4 ± 4.0	64.2 ± 2.7

6.4 Joint angles, prototypical classification, and crops

Table 7 and Figure 8 report the follow-on matrix on the same 196 clips per seed. Concatenating the 20 angles under cross entropy raises the BiLSTM from 24.8% ± 4.0% to 37.9% ± 2.4%. Consonant accuracy moves from 10.0% to 28.7% ± 4.5%; vowels stay near 63.5%. The angle MLP does not beat the xyz BiLSTM.

Prototypical training is the larger change. An xyz prototypical BiLSTM reaches 55.8% ± 1.8% accuracy and 53.4%±2.8% macro-F1. Adding angles to that encoder reaches 59.4%±3.3% accuracy and 57.4% ± 3.6% macro-F1. Consonants are 61.8% ± 5.7%; vowels are 52.6% ± 4.8%. The softmax xyz model was a vowel specialist. The prototypical encoder trades some of that vowel rate for a usable consonant rate. Fold means are 63.3% ± 3.7% on S1 and 55.4% ± 3.3% on S2.

The cropped-hand CNN is 19.2% ± 1.3%. Cropping beats full-frame RGB (5.8%–7.0%) and still loses to landmarks. Equal-weight softmax fusion of the prototypical encoder and the crop CNN is 59.7% ± 3.6%, within noise of the prototypical encoder alone. Fusion of the angle BiLSTM and the crop CNN is 41.5% ± 0.8%, a few points above that BiLSTM and well below prototypical classification. Figure 9 is the prototypical confusion matrix.

6.5 Closed 1-shot enrollment

Table 8 and Figure 10 separate train-only prototypes from a tutor setting where the system has already seen the test signer's other lighting of every letter. Closed 1-shot accuracy is 76.2% ± 4.2% for the xyz BiLSTM, 78.2% ± 2.8% for the angle BiLSTM, and 73.0% ± 1.5% for the prototypical encoder. Enrollment helps the weaker softmax models more. A new user who has not recorded those 49 clips still faces the 59.4% unseen-signer number.

Table 6: Batch-one clip latency. RGB end-to-end includes original-video decode on Apple M4 MPS. Landmark end-to-end is MediaPipe Hands on 16 cached frames plus the CPU classifier. Streaming FPS is reported separately.

Model	Device	Model-only (ms)	End-to-end (ms)	Clips/s
Single-frame CNN	MPS	4.0 (4.2)	939.0 (2097.8)	1.06
CNN–BiLSTM	MPS	49.5 (50.2)	962.9 (2126.2)	1.04
CNN–Transformer	MPS	49.7 (50.1)	970.9 (2162.6)	1.03
Landmark BiLSTM	CPU	0.3 (0.5)	345.7 (441.2)	2.89

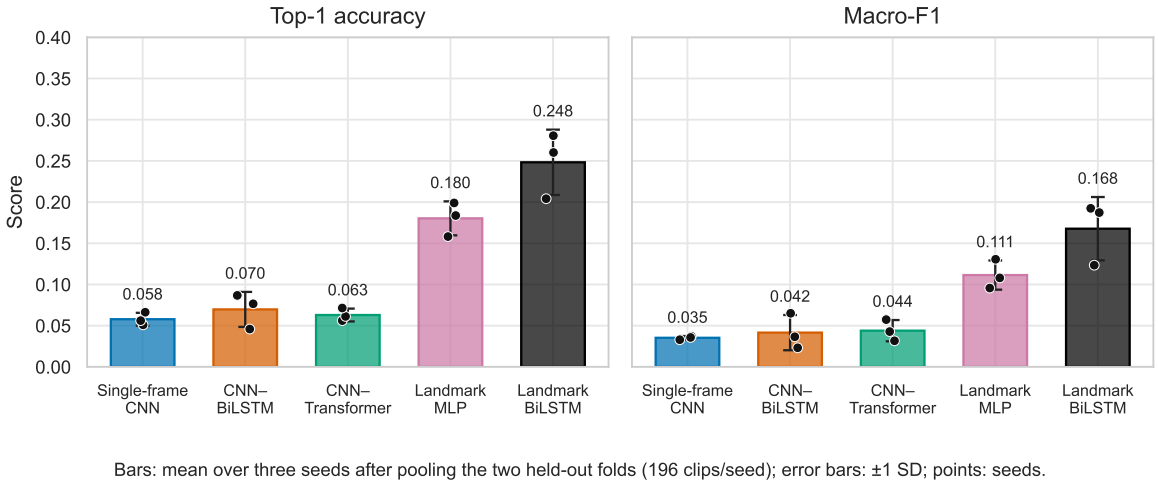


Figure 3: Pooled cross-signer development accuracy and macro-F1. Bars are means over three seeds; error bars are one sample standard deviation; points are seeds.

6.6 Other grading conventions

Table 9 and Figure 11 keep the encoder fixed and change the exam. High numbers in other NSL and few-shot papers often come from this kind of change: random image splits, fewer classes, or N-way episodes.

Top-3 and top-5 ask whether the true class is among the nearest three or five prototypes ($85.5\% \pm 6.0\%$ and $91.3\% \pm 4.0\%$). Random chance for those hits is 6.1% and 10.2%. Scoring only the 36 consonants, still unseen-signer, gives $65.7\% \pm 3.6\%$ (chance 2.8%). A 5-way 1-shot protocol, 500 episodes per run, reaches $90.6\% \pm 2.5\%$ with unseen-signer training prototypes and $93.3\% \pm 1.5\%$ when the support clip is the other lighting of the same person (chance 20%). Those 5-way figures belong next to Chamachot and Lertniphonphan's tables [2]. Quoting 91% as isolated NSL23 letter accuracy would mix those exams with the 49-way test.

6.7 Runtime and webcam

Table 6 separates network time from pipeline time. On MPS, RGB model-only medians are 4.0 ms (single-frame) and about 50 ms (temporal heads). End-to-end RGB clip processing is about 0.94–0.97 s because decoding dominates. A camera frame rate would have to count frames from a live stream, which this column does not.

On CPU, the landmark BiLSTM forward pass is 0.3 ms per 16-frame clip (0.5 ms at the 95th

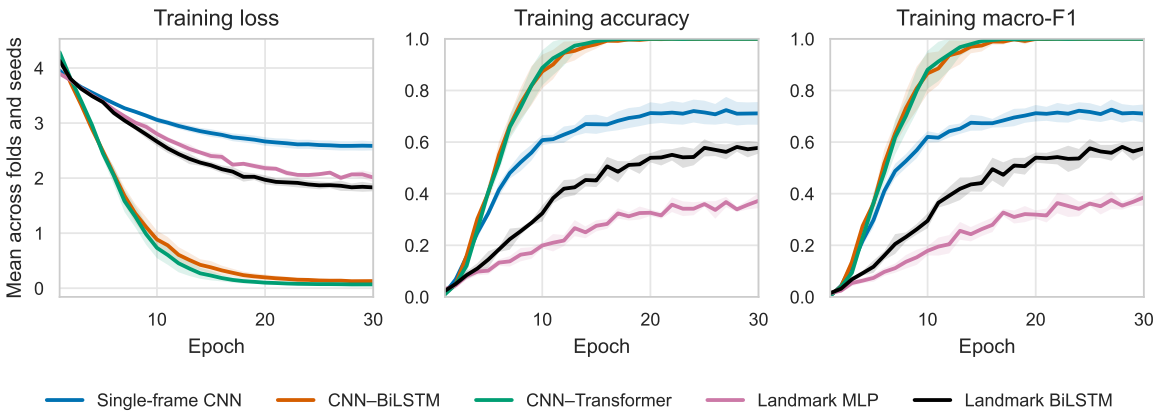


Figure 4: Training loss, accuracy, and macro-F1 (mean ± SD over folds and seeds). RGB temporal heads fit the training clips; landmark models do not reach training accuracy 1.

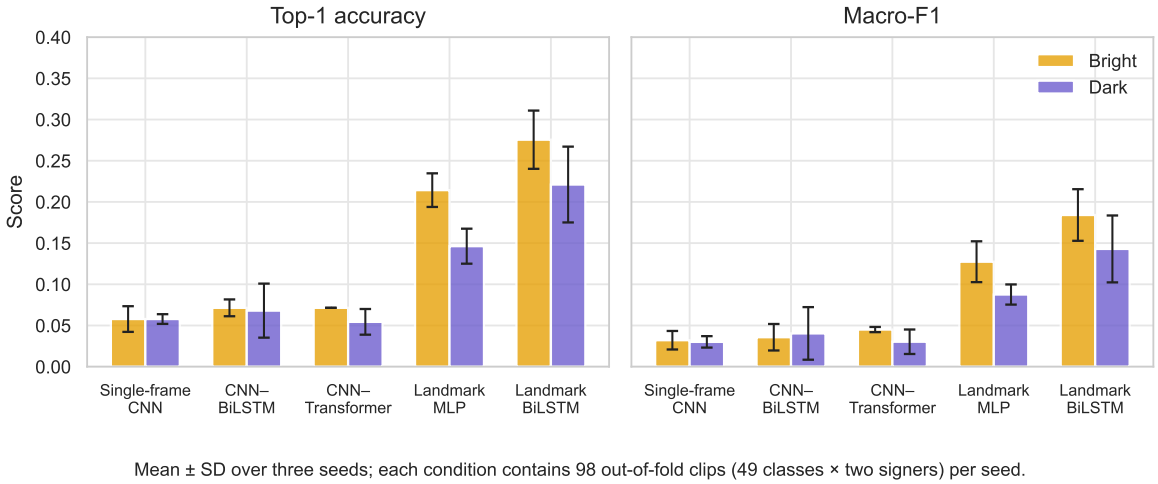


Figure 5: Bright versus dark accuracy and macro-F1. Each condition has 98 out-of-fold clips per seed.

percentile). MediaPipe on those 16 cached frames plus the classifier takes a median 346 ms per clip. The webcam loop is cheaper per frame because Hands runs in video mode on one frame at a time and the LSTM sees a rolling window.

On original NSL23 videos, with the prototypical xyz+angle encoder trained without the signer in the file, the loop processed 90 frames of S1's bright isolated `A` in 2.27 s (39.6 FPS). The argmax after EMA smoothing was `A`. The earlier xyz softmax loop on the same clip was 39.7 FPS; nearest-prototype classification leaves the MediaPipe-dominated budget almost unchanged. Three earlier xyz-softmax streams (S1 bright, S1 dark, S2 bright `A`) ran at 33–42 FPS. That is a speed check on one vowel.

7 Discussion

The frozen RGB experiment is mostly a negative result. With ImageNet MobileNetV2 frozen and 16 letterboxed frames, a BiLSTM or Transformer head barely moves cross-signer accuracy.

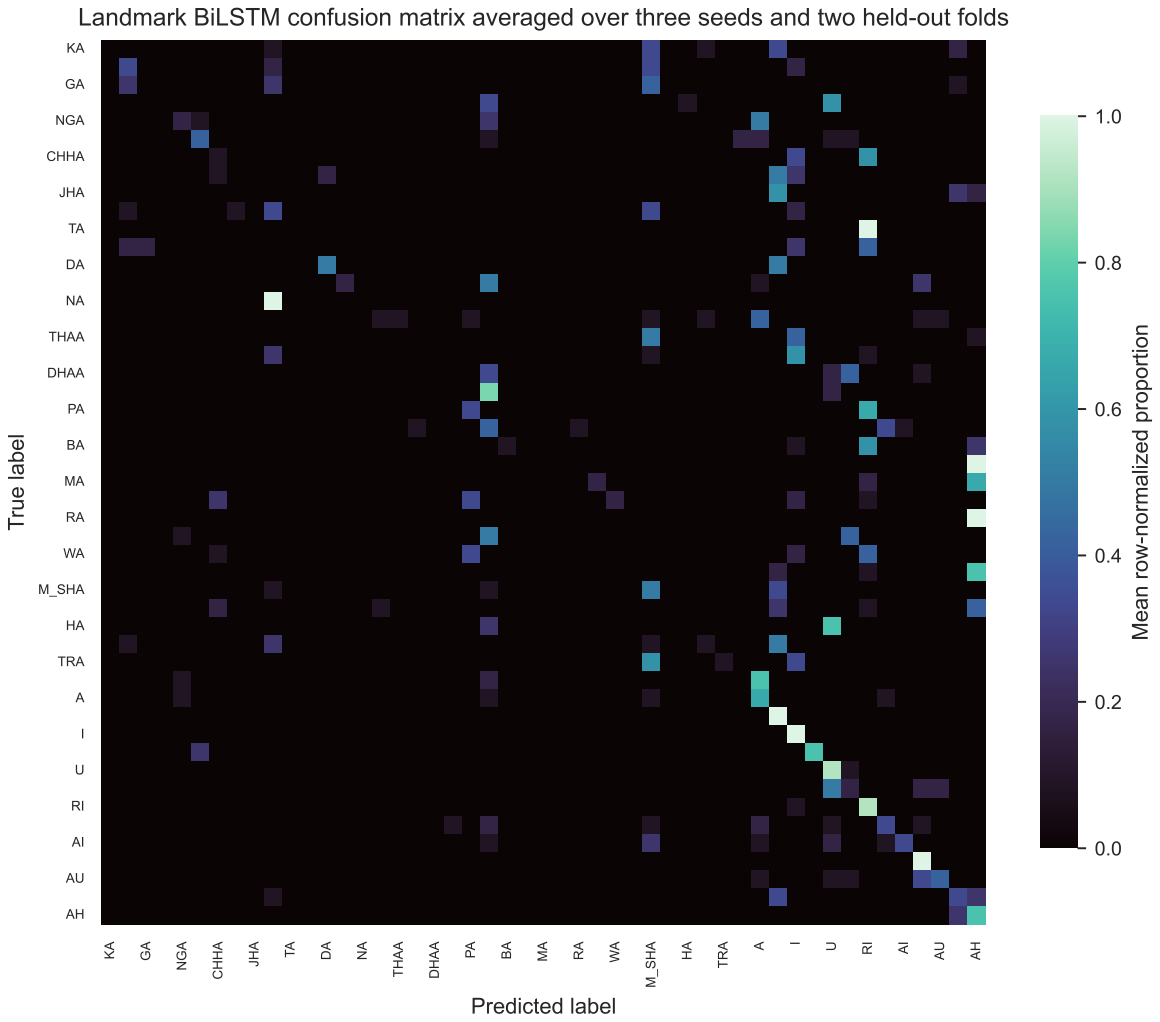


Figure 6: Landmark BiLSTM confusion, row-normalized and averaged over two folds and three seeds. Several consonants collapse onto a few vowel-like predictions.

Training accuracy hits 1.00 while evaluation stays near 6%. Extra temporal capacity is fitting the six people in the training fold. Many NSL23 consonants are held shapes; ImageNet features of a letterboxed room are a poor description of those shapes once the person changes.

Switching the input to MediaPipe coordinates is the first real gain. The 64-D wrist-relative vector discards most of the shirt and wall. Under ordinary softmax, the 73k-parameter BiLSTM at 24.8% is the control we keep quoting. Vowels carry almost all of that number (66% versus 10% on consonants). BHA, RA, and TA never come back correctly across twelve evaluations. That is a bad place to stop if the product is an alphabet tutor.

Angles plus prototypical loss are what move the consonants. The exploratory cross-signer estimate of 59.4% is still roughly two mistakes in five letters, so it is a weak tutor for a new person. It is also the first setting in this development study where the 36 consonants sit above 60%. Closed 1-shot on the other lighting of the same session is 76–78% for the softmax models; that number assumes the learner already recorded a dark and a bright take of every letter. Mixing the two numbers in a demo caption would overstate what a first-time user gets.

Cropping the hand for the same frozen CNN reaches 19.2%. The hand was small in the

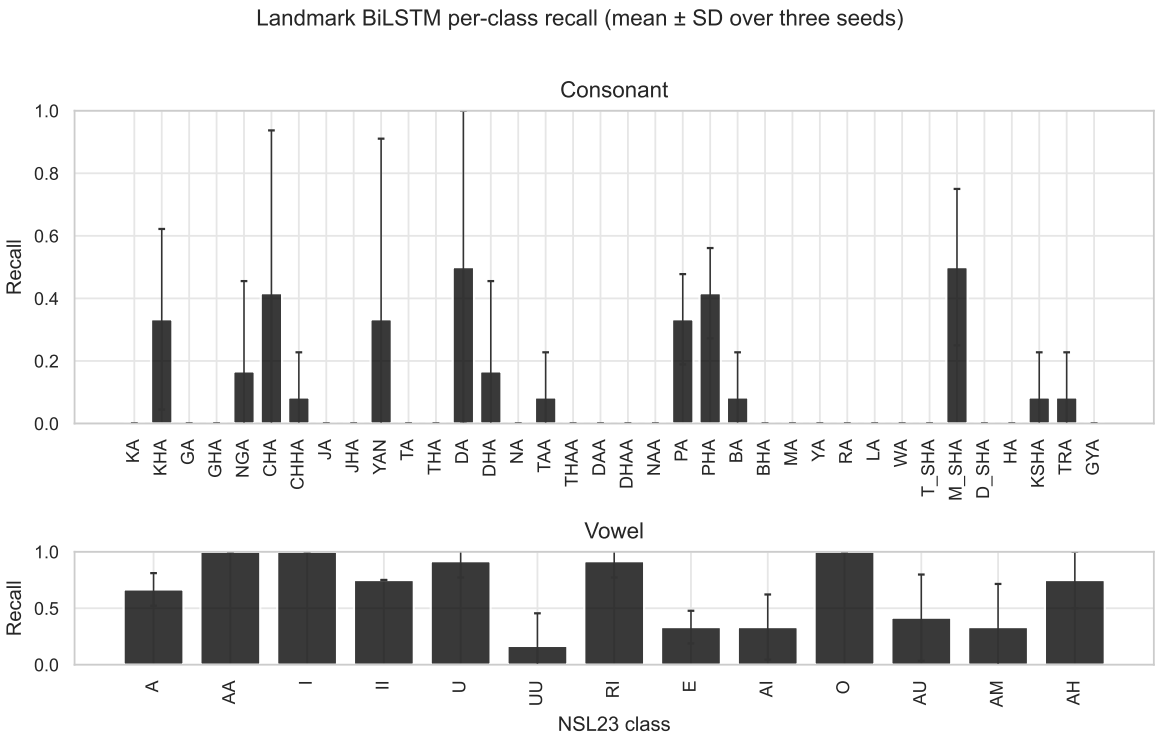


Figure 7: Per-class recall of the landmark BiLSTM (mean ± SD over seeds). Vowels occupy the lower panel.

Table 7: Follow-on unseen-signer accuracy on the same S1/S2 folds (196 clips/seed; mean ± SD over three seeds). The xyz BiLSTM row is the frozen 30-run control.

Model	Acc. (%)	Macro-F1 (%)	Cons. (%)	Vow. (%)
xyz BiLSTM (CE)	24.8 ± 4.0	16.8 ± 3.8	10.0 ± 3.9	66.0 ± 4.0
Hand-crop CNN	19.2 ± 1.3	16.9 ± 0.9	15.5 ± 0.8	29.5 ± 2.9
xyz+angle MLP (CE)	22.4 ± 3.1	15.6 ± 2.7	13.7 ± 3.6	46.8 ± 2.9
xyz+angle BiLSTM (CE)	37.9 ± 2.4	31.5 ± 3.4	28.7 ± 4.5	63.5 ± 3.8
xyz Proto-BiLSTM	55.8 ± 1.8	53.4 ± 2.8	58.3 ± 2.5	48.7 ± 5.9
xyz+angle Proto-BiLSTM	59.4 ± 3.3	57.4 ± 3.6	61.8 ± 5.7	52.6 ± 4.8

letterbox, but that is only part of the RGB failure. The encoder still describes these letter shapes poorly across people.

Percentages from other NSL papers stay on their own tables. Poudel-style still-image splits, Heuju's nine-word subset, and Chamachot's 5-way episodes answer different questions. The 90.6% 5-way figure is in Table 9 so that comparison is possible. The defensible Version 1 statement is an exploratory 59.4% cross-signer development estimate on these fixed clips, not a confirmatory estimate for future signers [2].

8 Limitations

S1 and S2 are the entire complete-alphabet cross-signer development set. They were held out within each training run, but they also informed the later angle, crop, prototypical, and fusion comparisons. Bootstrap intervals on 196 clips describe those clips; they neither restore an

Table 8: Closed 1-shot enrollment: the other lighting of the held-out signer supplies one support clip per class (98 queries/fold). Mean ± SD over three seeds after pooling S1 and S2.

Encoder	Unseen signer (%)	Closed 1-shot (%)
xyz BiLSTM (CE)	24.8 ± 4.0	76.2 ± 4.2
xyz+angle BiLSTM (CE)	37.9 ± 2.4	78.2 ± 2.8
xyz+angle Proto-BiLSTM	59.4 ± 3.3	73.0 ± 1.5

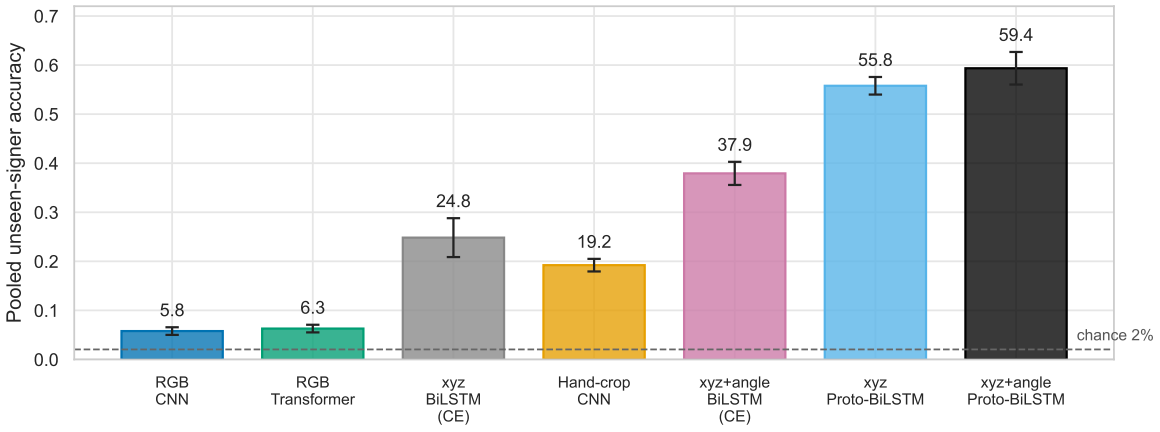


Figure 8: Unseen-signer accuracy after pooling S1 and S2. The xyz BiLSTM is the frozen 30-run control; the remaining bars are the follow-on matrix.

untouched test nor describe a population of signers. The five prepared-consonant people never sit in the complete-alphabet evaluation seat. Outdoor and real-world files in the Zenodo dump were left out of the primary sample because they fail the original-isolated-labelled audit. The clips themselves are isolated one-handed letters; sentence-level fingerspelling and body-relative signs are absent.

MediaPipe Hands is someone else's detector. Detection on these 6,000 cached frames was 97.8%, with no empty clips; a new webcam, darker rooms, or different skin tones can miss more. We did not swap in another tracker after seeing errors.

The RGB encoder stayed frozen on purpose. Fine-tuning MobileNetV2, or pretraining on a still-image letter set, might lift the crop CNN. Poudel's Kaggle consonants were unused: this machine had no Kaggle credentials, and that set is 36 classes rather than 49. Hyperparameters were locked before the RGB matrix; they may be suboptimal.

Enrollment here is the other lighting of the same session. There is no second-day recording and no live enrollment UI. Two clips per class per signer also means 5-shot enrollment of the test person has no honest support set.

The FPS numbers come from replaying NSL23 `.MOV` files on an Apple M4 with MediaPipe 0.10.14 on CPU. A live user study would be a different paper. Interactive speed and recognition quality are separate measurements.

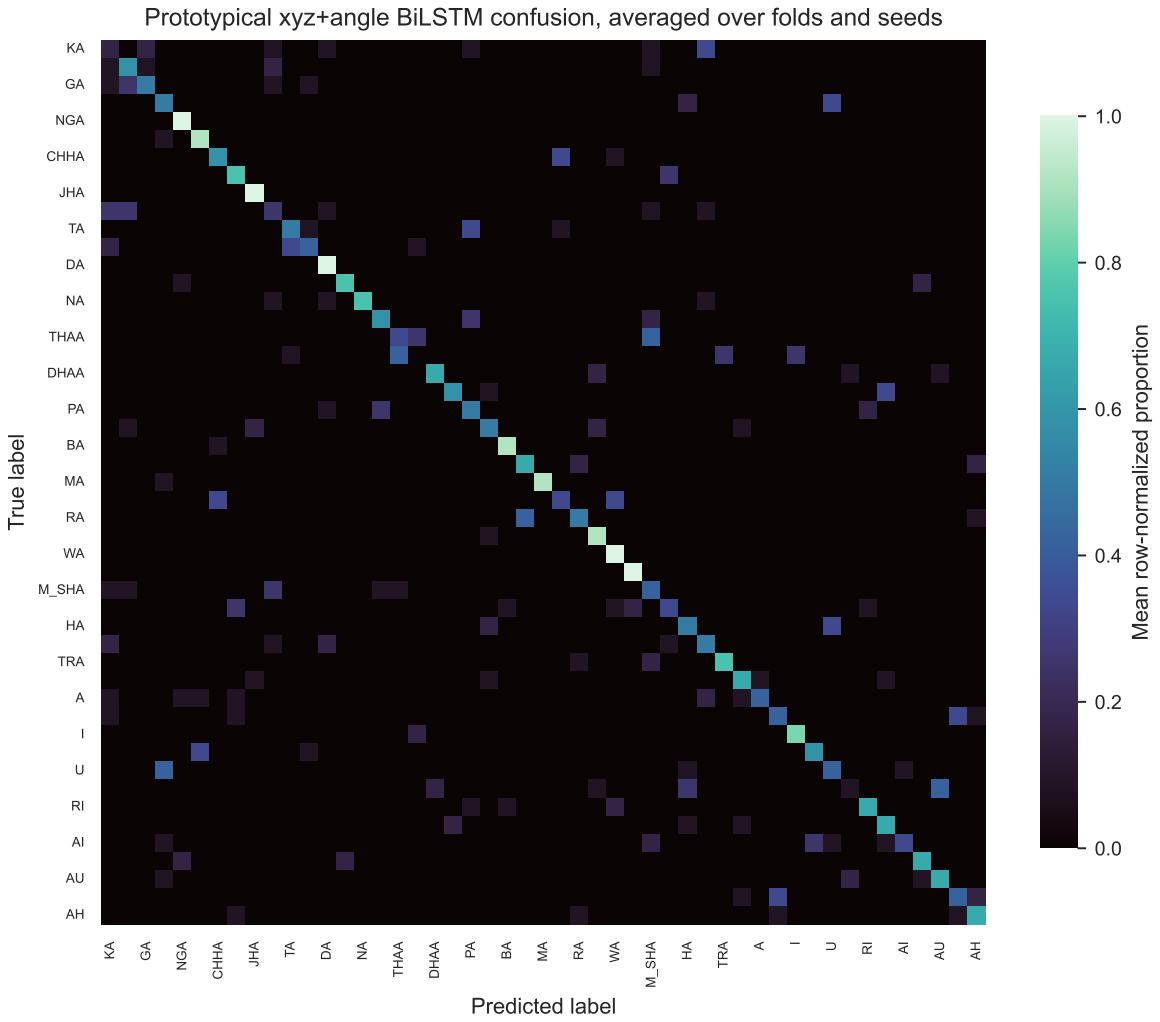


Figure 9: Prototypical xyz+angle BiLSTM confusion, row-normalized and averaged over two folds and three seeds.

9 Conclusion

For a webcam tutor, the exploratory cross-signer development estimate on this sample is 59.4%± 3.3% on 375 audited isolated NSL23 clips, with S1 and S2 held out within each run. Frozen RGB never left the 6–7% band. Wrist-normalized landmarks, 20 joint angles, and prototypical classification are what moved consonants to 61.8%. A CPU loop on an Apple M4 keeps up with the camera at 33–42 FPS. Reusing two complete-alphabet people for follow-on development is too thin for a deployment or population claim, including in rural Nepal. A prospective evaluation needs newly frozen signers; later work still needs in-country data. Code, folds, and predictions are in the project repository so the sample definition can stay fixed while the detector or encoder changes.

Table 9: The same xyz+angle prototypical encoder under exams used in other sign and few-shot papers. Primary claim is still 49-way unseen top-1. Five-way episodes use 500 draws per run.

Exam	Chance (%)	Score (%)
49-way top-1, unseen signer (primary)	2.0	59.4 ± 3.3
49-way top-3, unseen signer	6.1	85.5 ± 6.0
49-way top-5, unseen signer	10.2	91.3 ± 4.0
36-way consonants, unseen signer	2.8	65.7 ± 3.6
5-way 1-shot, unseen-signer prototypes	20.0	90.6 ± 2.5
5-way 1-shot, same signer, other lighting	20.0	93.3 ± 1.5
49-way closed 1-shot, other lighting	2.0	73.0 ± 1.5

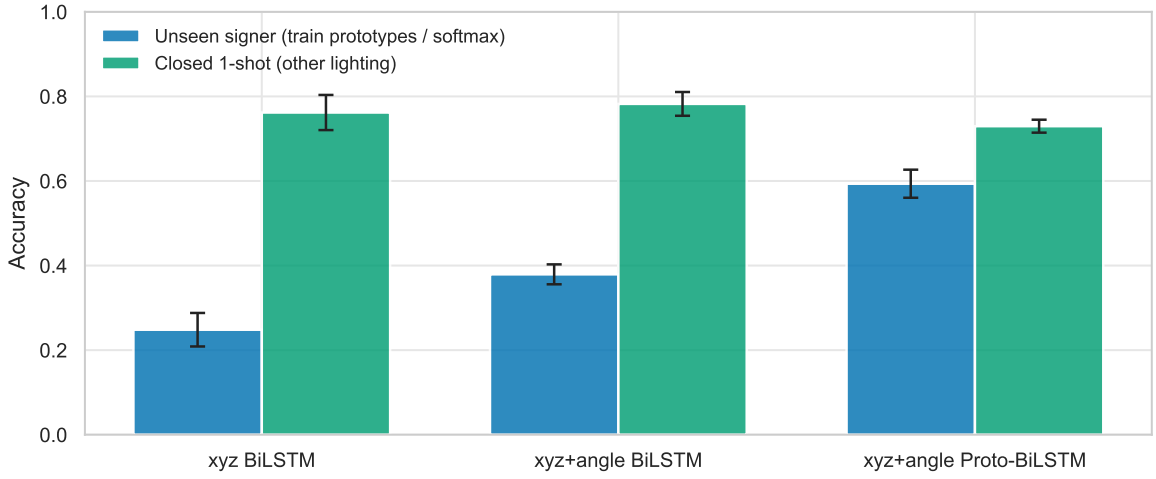


Figure 10: Unseen-signer classification versus closed 1-shot enrollment on the other lighting of the held-out signer.

Data availability

NSL23 videos are distributed by Sunuwar, Borah, and Kharga under CC BY 4.0 [15, 16]. This repository does not republish the archives. Code, portable fold manifests, landmark caches, checkpoints, tabulated results, the immutable Version 1 snapshot, its historical-tree verifier, and the Version 2 annotation protocol are in the project repository.

Acknowledgements

The recordings were collected and released by the NSL23 authors. This paper is a secondary analysis of those files.

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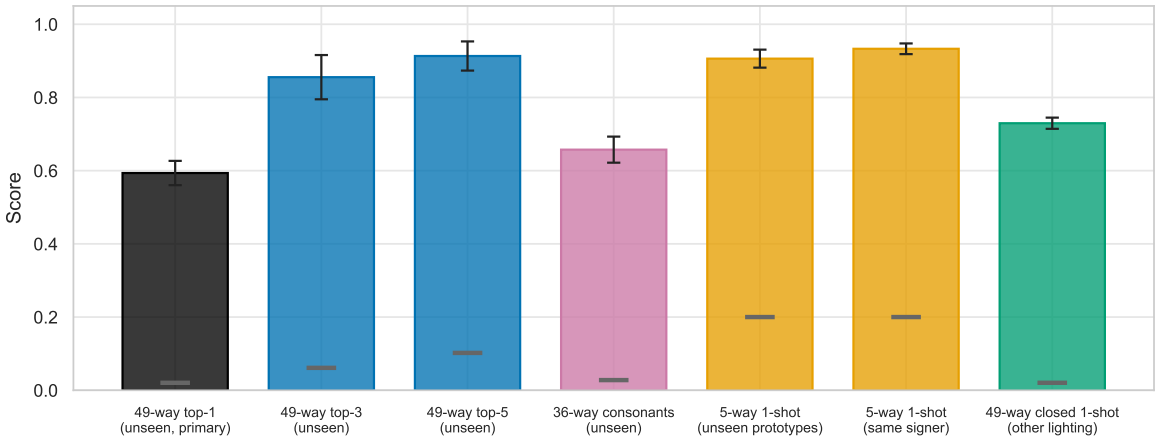


Figure 11: Same prototypical encoder, different exams. Grey marks are chance. The leftmost bar is the primary 49-way unseen top-1 claim.

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